



Effect of planar torsional deformation on the thermal conductivity of 2D nanomaterials: A molecular dynamics study

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ARTICLE INFO

Keywords:

Thermal conductivity
2D materials
Torsion
Molecular Dynamics

ABSTRACT

The thermal conductivity of nanoscale materials is largely dependent on the applied strain and deformations. In this paper, the effect of torsional deformation and consequent wrinkles on the thermal conductivity of graphene, hexagonal boron nitride (h-BN) and molybdenum disulfide (MoS_2) nanostructures have been investigated by performing non-equilibrium molecular dynamics simulation. The nanostructure geometry is considered as a ring with different inner and outer radii so that the torsional deformation is applied on the inner boundary. It is found that the wrinkles caused by applying the torsion, result in reducing the thermal conductivity of nanostructures. Although the effect of created distortions is tangible, these wrinkles have the most influence on the thermal conductivity of MoS_2 and the least on the thermal conductivity of h-BN. The effect of inner radius size is also studied and found that the reduction of the thermal conductivity is enhanced by increasing the inner radius size. The results of this study can be beneficial to thermal applications of 2D nanostructures under mechanical stress or strain and also for estimating the values of applied torsion and wrinkle amplitude by measuring the thermal conductivity variations.

1. Introduction

Inspired by the graphene exploration and its sole characteristics [1], research on one atom thick two-dimensional nanostructures quickly developed in the scientific and engineering subfields. In recent years, due to possible applications in nanoelectronics, two-dimensional materials are one of the most important fields of research in nanomaterials [2–4]. Over the past decade, numerous studies have been done on the study of the properties of two-dimensional nanostructures especially in engineering applications such as nanosensors, energy storage devices and solar cells [5–7]. Due to the low thickness and slight bending strength of two-dimensional structures, some wrinkles are formed in these materials [8,9]. Although the most significant factor in forming wrinkles is inherent properties but heating and cooling operations, mechanical strains, and boundary conditions are among the factors that affect the accumulation and amplitude of these wrinkles [10–14]. These out-of-plane deformations can have both positive and negative effects on the properties of nanomaterials. The study of wrinkles in two-dimensional nanostructures has been limited to the wrinkling mechanism [15,16] and their effect on the electronic transport properties. It has been shown that multiplicity of chemical carbon-carbon bonds leads to unintentional ripples in graphene plate that appear due to thermal fluctuations [17]. In addition to thermal fluctuations, interatomic forces

between the nanostructures and substrate can also contribute to generation and amplification of ripples [18]. In recent years, some studies have been evolved on the created pattern as a result of the growth of a thin elastic layer in macro scale [19–21]. Moreover, further studies have been done on similar behaviors, like bending and buckling on carbon nanotubes [22,23].

In order to tune the thermal conductivity of nanostructures, different methods such as defect-inducing [24–28], applying stress and strain [29–32], impurity-doping [33–35], and nano-coating [36] have been used to engineer the properties of two-dimensional nanostructures. The application of the torsional deformation and the resulting distortions can also be one of the methods to achieve desired properties. Previous studies have shown that the formation of wrinkles in graphene increases the phonon dispersion and consequently decreases the thermal conductivity of graphene [37–39]. Furthermore, these distortions can lead to anisotropy in the thermal conductivity of graphene [40]. According to studies on the thermal properties of graphene, the behavior prediction of other two-dimensional nanostructures such as MoS_2 and h-BN in the presence of wrinkles is also a challenging subject.

In this paper, we use non-equilibrium molecular dynamics (MD) simulations to study the effect of torsional deformation and its resulting wrinkles on the thermal conductivities of graphene, h-BN, and MoS_2 nanostructures. The thermal conductivity is calculated by applying a

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<https://doi.org/10.1016/j.mtcomm.2019.100706>

Received 18 September 2019; Received in revised form 15 October 2019; Accepted 15 October 2019

Available online 24 October 2019

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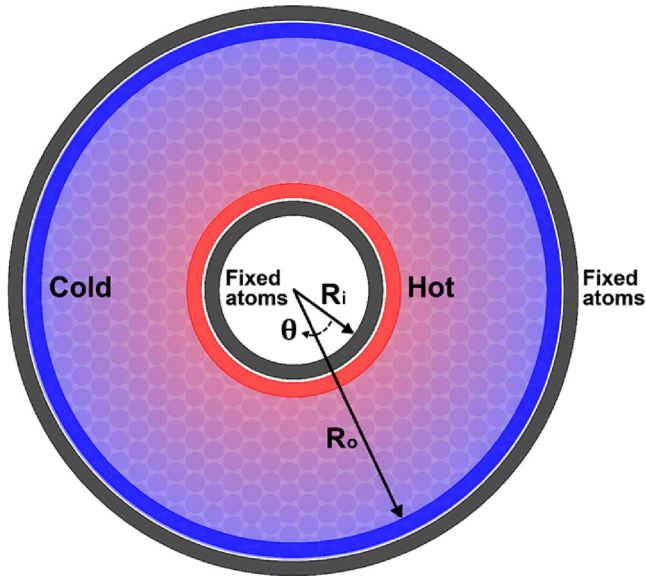


Fig. 1. MD simulation setup for thermal conductivity prediction of graphene, MoS₂, and h-BN under torsional deformation (θ) applied on the inner boundary.

temperature gradient in the radial direction. After validation of MD results for pristine forms of the structures, the thermal conductivities are calculated at different torsional angles up to the failure points for each structure. The relation between the thermal conductivity reduction and the wrinkle amplitude as well as the effects of geometry and number of layers are also discussed.

2. Molecular dynamics simulations

The thermal conductivities of structures are calculated by applying a non-equilibrium molecular dynamics method [41] utilizing the LAMMPS package [42]. The optimized Tersoff potential function [43] has been employed to describe C-C and B-N interatomic interactions in graphene and h-BN nanostructures, respectively. To simulate the structure of MoS₂, the REBO potential is used as well [44]. A schematic view of the atomistic model with an internal radius $R_i = 7$ nm and an external radius $R_o = 25$ nm is shown in Fig. 1.

In Fig. 1, the gray boundaries (innermost and outermost boundaries) represent the internal and external boundary conditions. Initially, the structures are relaxed and equilibrated at $T = 300$ K by performing the Nosé-Hoover barostat and thermostat (NPT) simulations over a period of 0.5 ns with a time increment of 0.1 fs. Afterward, the microcanonical ensemble (NVE) and then the boundary conditions are applied to the system. First, the outer part of the exterior boundary of the plate is fixed, and then the boundary conditions $T_H = 320$ K and $T_C = 280$ K are applied to hot and cold rings adjacent to the internal and external boundaries, respectively. After applying the temperature boundary conditions, the internal boundary is rotated at an angular velocity of 0.1 rad.ps^{-1} to reach the desired torsional deformation level.

To calculate the temperature gradient, the simulation box is apportioned into 20 slabs along the radial direction and the temperature at each slab is determined separately. The simulation is carried out for 10 ns with NVE ensemble. As the average temperature at each slab is calculated, a logarithmic temperature distribution appeared in the middle of the specimen as shown in Fig. 2a. In this method, the heat flux q can be calculated from the energy exchange between cold and hot boundaries as shown in Fig. 2b.

Fig. 2a shows the graphene temperature profile in the radial direction. For the steady-state condition with radial symmetry, the heat equation is:

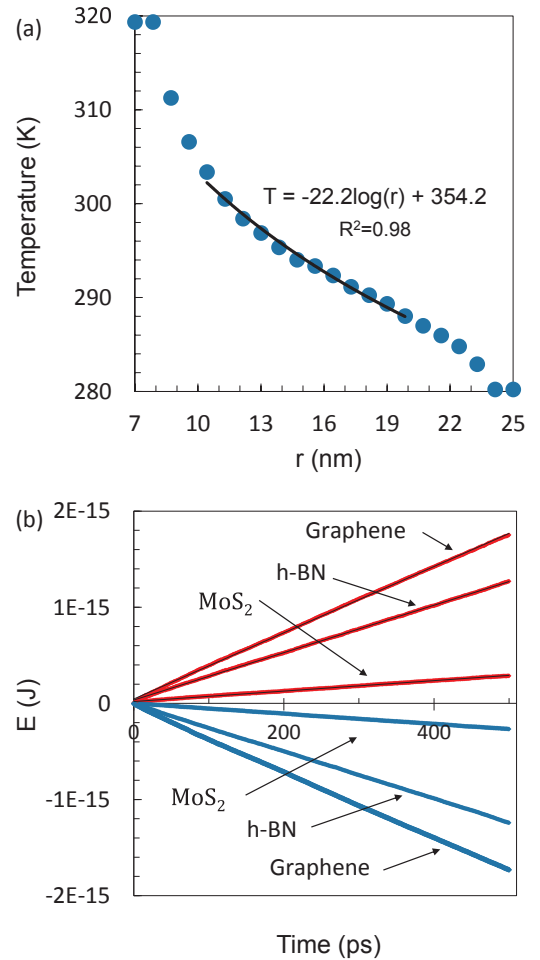


Fig. 2. (a) Temperature gradient obtained from NEMD simulation to calculate $\frac{\partial T}{\partial r}$. The logarithmic curve was fitted according to Eq. (2) (b) Added energy to the hot ring and subtracted energy from the cold ring with respect to the simulation time. The slopes (3 mW, 2 mW and 0.5 mW for graphene, h-BN and MoS₂, respectively) are equal to the heat flow in each system. Both figures were calculated for pristine forms of the nanostructures.

$$\nabla^2 T = \frac{1}{r} \frac{\partial T}{\partial r} + \frac{\partial^2 T}{\partial r^2} = 0, \text{ with } T|_{r=R_i} = T_H, T|_{r=R_o} = T_C \quad (1)$$

Eq. 1 has the analytical solution for the temperature profile:

$$T(r) = T_H + (T_C - T_H) \left(\log \frac{r}{R_i} \right) / \log \frac{R_o}{R_i} \quad (2)$$

The effective thermal conductivity is then calculated by Fourier's law.

$$\kappa = -q'' \left(\frac{\partial T}{\partial r} \right)^{-1}$$

where q'' is the heat flux and $\frac{\partial T}{\partial r}$ is the temperature gradient in the system, which are calculated from Fig. 2.

3. Results and discussions

3.1. Validation

To validate our MD results, the thermal conductivities of graphene, h-BN and MoS₂ are calculated without applying torsional deformation. As shown in Table 1, graphene has a thermal conductivity close to the previously reported value for the graphene sheet [45]. Results for h-BN and MoS₂ also show less than 10% of variation with values of previous

Table 1

Comparison of the thermal conductivity calculated in this study and previous studies.

Material	Length (nm)	Present study	Previous researches
Graphene	18	199.2 ± 2.7	183.7 [45]
MoS ₂	20	5.4 ± 0.1	5.9 [47]
h-BN	50	69.1 ± 3.8	66 ± 2 [46]

studies [46,47]. It should be noted that the experimental values for thermal conductivity of two-dimensional nanostructures are higher than the calculated values in the NEMD method. For example, graphene has a thermal conductivity about 5300 W/mK [48], much higher than the computationally reported values. This disagreement is related to the size-dependency in thermal properties of two-dimensional nanostructures, a phenomenon which has been noted by other researchers [49–51]. Two-dimensional samples used in laboratories are usually a few micrometers long; However, given the high computational cost to simulate such large specimens by non-equilibrium molecular dynamics, and given that the radius used in this study is less than the average mean free path of these nanostructures, here the qualitative study of the thermal conductivity behavior has been studied by applying plane strain.

3.2. Effect of torsional deformation

In this study, we investigate the thermal conductivity of three types of 2D nanostructures under twisting at the internal boundary. We first define shear strain (γ) parameter with respect to the torsional deformation (θ) as follows:

$$\gamma = \left(\frac{R_i}{R_o - R_i} \right) * \theta \quad (3)$$

It is found that rupture occurs in all types of 2D nanostructures at a similar strain around $\gamma \approx 0.07$ ($\theta = 11^\circ$). It should be noted that the value

of θ must be in radian in Eq. 3.

Fig. 3 shows the structural morphology of graphene, h-BN, and MoS₂ at $\gamma = 0$. It can be seen that they have random fluctuations in their height (z-direction). These random fluctuations emerge due to the 2D nature of these nanostructures [8,9]. After the strain is applied to the internal boundary of these structures, regular forms of wrinkles begin to reveal. In Fig. 3, the amplitude and morphology of the created wrinkles at $\gamma = 0$ and $\gamma \approx 0.03$ (equivalent to $\theta = 0^\circ$ and $\theta = 5^\circ$, respectively) are demonstrated. As shown, the wrinkles are randomly distributed at $\gamma = 0$, but the torsional deformation induces regular wrinkles in these nanostructures. The amplitude of the wrinkles in the MoS₂ nanostructures at $\gamma \approx 0.03$ reaches up to 1 nm. This amplitude varies in each nanostructure.

Fig. 4a shows the thermal conductivity of 2D nanostructures with respect to the applied strain, normalized by the thermal conductivity of pristine form. By increasing the strain (torsional deformation), the thermal conductivity of all 2D nanostructures decreases. By applying torsional strain, elongation and compression happen within the atomic bonds of two-dimensional nanostructures. This change in the bond's length causes minor variations in the vibrational patterns and thus reduces thermal conductivity [52]. Moreover, the wrinkles increase the phonon scattering rates in a deformed 2D nanostructure that leads to a decrease in the thermal conductivity [30]. It is also found that the most reduction in the thermal conductivity occurs for MoS₂ nanostructure; while the least belongs to h-BN nanostructure. The reduction of thermal conductivity is related to the amplitude of wrinkles as shown in Fig. 4b. The highest amplitude of the wrinkles in the investigated nanostructures is for MoS₂ which was also shown in Fig. 3.

To investigate the relation between the wrinkle's amplitudes and the reduction of the thermal conductivity of a 2D material, we present the thermal conductivity in terms of wrinkle amplitude of graphene nanostructure under different values of the torsional deformation (from $\theta = 0^\circ$ to the failure point) in Fig. 5. It is found that as the wrinkle amplitude increases, the thermal conductivity will be further reduced. Fig. 5 shows that the thermal conductivity reduction is linearly

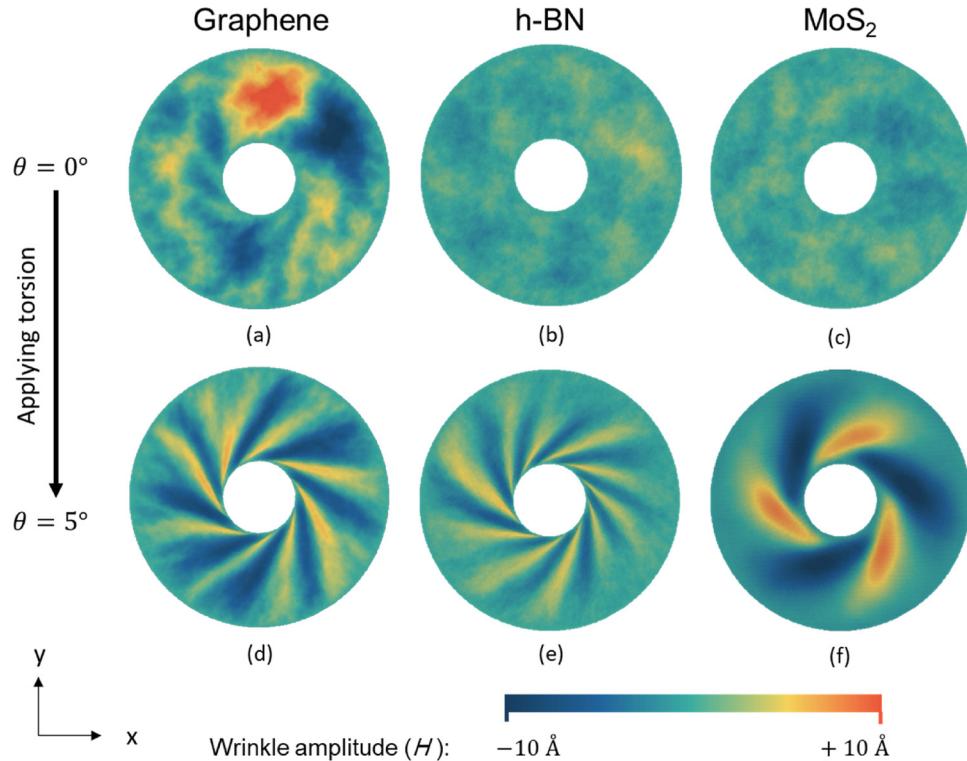


Fig. 3. Contours of wrinkles amplitude in two-dimensional nanostructures (graphene, h-BN, MoS₂) in pristine forms (a, b, c) and under torsional deformations $\theta = 5^\circ$ (d, e, f).

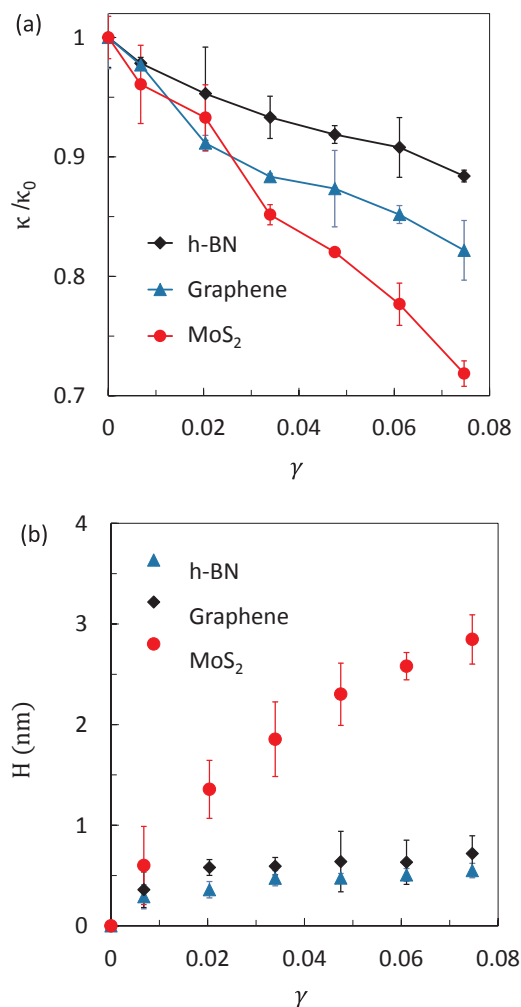


Fig. 4. (a) In-plane thermal conductivities of graphene, h-BN and MoS₂ under torsional deformations with respect to the applied strain. Results were normalized with the thermal conductivities of pristine forms (κ_0). (b) Wrinkles amplitude of structures under torsional deformations.

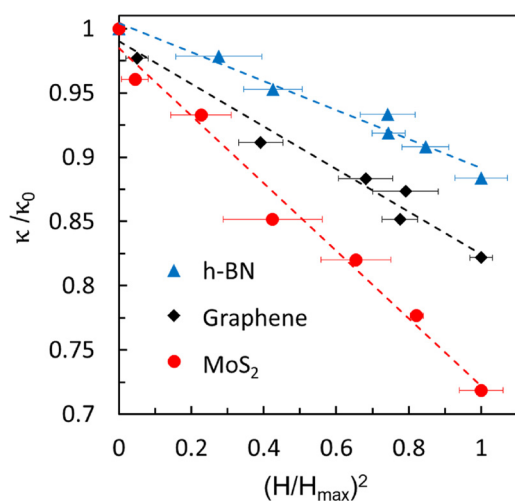


Fig. 5. The thermal conductivity reduction versus of the square wrinkle amplitude (H) normalized by the wrinkle amplitude at the maximum strain level ($H_{\max}$).

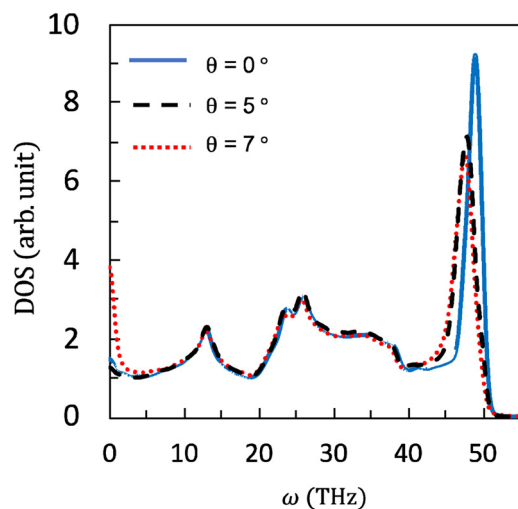


Fig. 6. Vibrational power spectra (PDOS) of graphene under torsional deformations. The PDOS peaks moves towards lower frequencies by applying the torsional deformation.

proportional to square of the wrinkle amplitude.

The vibrational power spectra (DOS) for graphene under various torsional deformation are shown in Fig. 6. The spectra are calculated for in-plane and out-of-plane modes through the Fourier's integral of the velocity auto-correlation function:

$$DOS(\omega) = \int_0^\infty \langle v(t)v(0) \rangle e^{-i\omega t} dt \quad (4)$$

where ω is the angular frequency and v is the velocity of atoms over simulation time.

Generally, the obtained phonon power spectra show phonons softening where the torsional deformation increases. Moving from high frequency peaks (G-band) toward lower frequencies, phonons velocity decreases which according to the classical heat transfer theories, it causes decreasing the thermal conductivity of the material.

3.3. Size effect

It has been shown that tuning wrinkling pattern is practical by manipulation of torsion direction and the size of the hole [53]. Due to this, to investigate the effect of geometry size on the thermal conductivity of structures under torsional deformation, the simulations have been performed on two graphene rings with different inner radii and the same outer radii. The thermal conductivity variations in two cases with radii $R_i = 7\text{nm}$ and $R_i = 10\text{nm}$ are shown in Fig. 7. It should be noted that the thermal conductivity of graphene strongly depends on the size of the system and as the size increases, the thermal conductivity will be further increase. Thus, the thermal conductivity of the second case ($R_i = 7\text{nm}$) in a pristine form is larger than that of the first case ($R_i = 10\text{nm}$).

To understand why the thermal conductivity decreases in larger inner radius, wrinkles amplitude contours are shown in Fig. 8. It is found that the number of wrinkles increases when the inner radius increase, which leads to increase in the phonon scattering rate within graphene and resulting to further reduction of the thermal conductivity.

3.4. Effect of number of layers

Nanostructures are usually produced in multilayer forms in experimental synthesis. So it is essential to investigate the effect of torsional strain on the thermal conductivity of few-layer nanostructures. For this purpose, monolayer to four-layer graphene sheets was studied under torsion and their thermal conductivities were calculated at a specific torsional value ($\theta = 5^\circ$). As can be seen in Fig. 9, the thermal

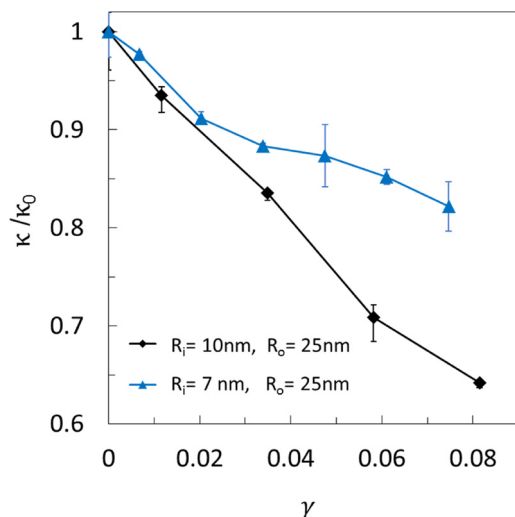


Fig. 7. The thermal conductivity of two graphene rings with different internal radii ($R_i = 7$ nm and $R_i = 10$ nm) and the same external radii ($R_o = 25$ nm) with respect to the torsional deformation.

conductivity reduction is not influenced by the increasing the number of layers. The reason may be the interplay of weak van der Waals interatomic interactions between the layers against the strongly covalent planar bonds.

4. Summary

In summary, utilizing non-equilibrium molecular dynamics method, it has been shown that wrinkles play an important role in the thermal conductivity of 2D nanostructures. The thermal conductivity of three types of 2D nanostructures (graphene, h-BN and MoS₂) were examined under torsional deformation. It was found that applying the torsional deformation up to the failure point results in a reduction of 20–30 percent in the thermal conductivity of these nanostructures. The reduction was the highest for MoS₂ nanostructure and the lowest for h-BN nanostructure. It was also revealed that the amplitude of wrinkles has a direct effect on the thermal conductivity of a 2D nanostructure, so that as the wrinkles amplitude (H) increases, the thermal conductivity (k) will be further reduced. These results have important implications for understanding heat transfer mechanisms in 2D materials and also

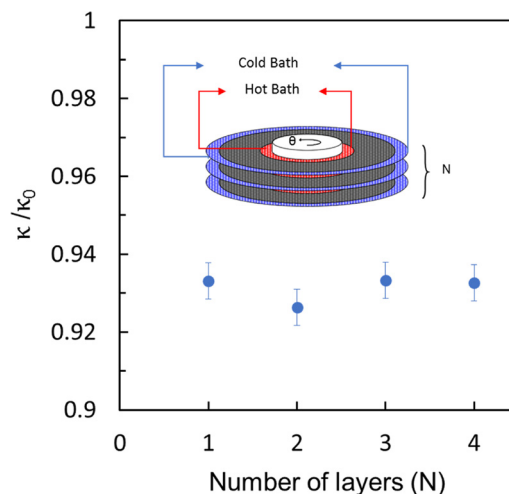


Fig. 9. In-plane thermal conductivity of few-layer graphene under torsional deformation ($\theta = 5^\circ$) versus the number of layers (N). Results were normalized with the thermal conductivities of pristine forms (κ_0).

for engineering of thermal transport in electronic devices.

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version. This manuscript has not been submitted to, nor is under review at, another journal or other publishing venue. The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript.

Declaration of Competing Interest

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version. This manuscript has not been submitted to, nor is

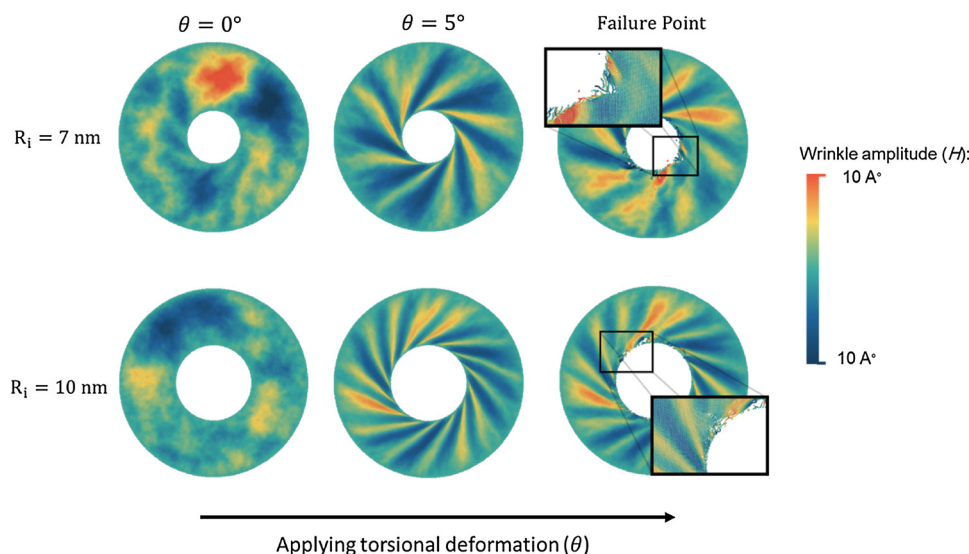


Fig. 8. Contours of wrinkle height in graphene rings with different inner radii as $R_i = 7$ nm and $R_i = 10$ nm under different torsional deformations ($\theta = 0^\circ$, $\theta = 5^\circ$ and failure point).

under review at, another journal or other publishing venue. The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript.

Acknowledgement

Authors greatly acknowledge professor Dominic Vella from university of Oxford for initiating the idea and revising the content of the paper.

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